MIL-C-46079A(MR)
28 June 1968
SUPERSEDING
MIL-C-46079(MR)
25 June 1964

MILITARY SPECIFICATION

COATING, EPOXY, BAKING TYPE FOR MAGNESIUM CASTINGS

1. SCOPE

1.1 This specification covers one type of single package unpigmented epoxy coating for sealing magnesium castings before painting.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-M-44 - Magnesium Alloy Plate and Sheet (AZ-31B).
TT-P-143 - Paint, Varnish, Lacquer and Related Materials; Packaging, Packing and Marking of.

MILITARY

MIL-M-3171 - Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion On.

STANDARDS

FEDERAL

Fed. Test Method Std. No. 141 - Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling and Testing.

FSC 8010

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:48Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Composition. The coating shall be furnished as a heat curable one-component clear liquid comprised of a blend of bisphenol type epoxy resins, curing agents and/or other additives dissolved in suitable solvents to yield a stable product conforming to the requirements of this specification. When tested as in 4.4.3 the nonvolatile portion of the coating shall exhibit only those absorption bands shown in figure 1 and the strength of these bands shall conform to the absorbance ratios in table III.

3.2 Quantitative requirements. The coating shall conform to the quantitative requirements specified in table I when tested as in 4.1.

Table I. Quantitative Requirements

Characteristics	Requirements	
	Minimum	Maximum
Total solids, percent by weight of coating	49	51
Epoxy resin, percent by weight of total solids	85	--
Viscosity, package, Gardner tubes	F	H
Viscosity, reduced, No. 4 Ford cup, seconds	10	15
Color, package, Gardner Color Std. 1953	--	8
Drying time		
Set to touch, minutes	--	5
Full hardness, after baking at 392° ± 4°F., minutes	--	70

3.3 Qualitative requirements.

3.3.1 Condition in container. When tested as in 4.4.6 the coating shall be clear, free from cloudiness, suspended matter, and sediment. A freshly opened container of the coating shall show no skinning, gelling or livering.

3.3.2 Accelerated storage stability. When tested as in 4.4.7, the coating shall be clear, free from cloudiness, suspended matter or sediment. The coating shall show no skinning, gelling or livering. The viscosity shall not be greater than Gardner tube K and the color not darker than Gardner Color Std. No. 12.

3.3.3 Dipping properties. When tested as in 4.4.8, the coating shall show satisfactory dipping properties. After final baking, the film shall be uniform, clear, hard, glossy and free from cloudiness, cratering, orange peel, silking or other film defects. When cut with a knife as in method 6304 of Fed. Test Method Std. No. 141 the film shall adhere tightly to the metal without chipping or flaking. The knife cut shall show beveled edges and no film separation resulting from poor intercoat adhesion when the cut is probed with the tip of the knife.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:48Z
Check the source to verify that this is the current version before use.

3.3.4 Spraying properties. When tested as in 4.4.9, the coating shall spray satisfactorily in all respects without running or sagging. After baking the film shall be uniform, clear, hard, glossy and free from cloudiness, cratering, orange peel, silking or other film defects.

3.3.5 Flexibility. When tested as in 4.4.10 the coating shall withstand bending without cracking, chipping, or flaking.

3.3.6 Salt spray resistance. When tested as in 4.4.11 the coating shall exhibit no blisters, undercutting or other film deterioration at the scribe mark before and after the 2 hour recovery period. All other areas of the panel shall be free of corrosion, blisters or other film defects of any kind.

3.3.7 Adhesion properties (wet). When tested as in 4.4.12 the coating shall not be removed or separated from the panel by the adhesive tape.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.2 Sampling, inspection and testing. Unless otherwise specified, sampling, inspection and testing shall be in accordance with section 1000 of Fed. Test Method Std. No. 141.

4.3 Classification of tests. Testing under this specification shall be for acceptance of individual lots.

4.4 Test methods.

4.4.1 Test conditions. The routine and referee testing conditions shall be in accordance with section 7, Fed. Test Method Std. No. 141 except as otherwise specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:48Z
Check the source to verify that this is the current version before use.

4.4.2 The following tests shall be conducted in accordance with Fed. Test Method Std. No. 141 except as hereinafter specified.

Table II. Index

Test	Test methods: Applicable method in Fed Std 141	Test methods: Paragraph of this specification giving further reference	Paragraph of this specification giving requirements
Vehicle identification, infrared analysis	--	4.4.3	3.1
Total solids	4041	--	Table I
Epoxy resin, chemical analysis	7401	--	Table I
Viscosity, package, Gardner tubes	4271	--	Table I
Viscosity, reduced, Ford cup	4282	4.4.4	Table I
Color, Gardner Color Std. 1953	4248	--	Table I
Drying time	4061	4.4.5	Table I
Condition in container	3011	4.4.6	3.3.1
Accelerated storage stability	3022	4.4.7	3.3.2
Dipping properties	4341	4.4.8	3.3.3
Spraying properties	4331	4.4.9	3.3.4
Flexibility	6221	4.4.10	3.3.5
Salt spray resistance	6061	4.4.11	3.3.6
Adhesion (wet) tape test	6301	4.4.12	3.3.7

4.4.3 Vehicle identification, infrared. Apply a suitable film of the packaged coating to a rock salt plate and dry in a vacuum oven at 70° ± 2°C. (158° ± 4°F.) for 30 minutes. Scan the infrared spectrum of the dried film using a rock salt plate as compensation in the reference beam. Compare the spectrum with that in figure 1 and calculate the absorbance ratios for compliance with 3.1, using the following equation:

$$A_r = \frac{A_\lambda - A_{2.55}}{A_{12.13} - A_{2.55}}$$

Where: A_r = absorbance ratio

A_λ = absorbance at the wave length in question

$A_{12.13}$ = absorbance of the band at 12.13 microns

$A_{2.55}$ = base line absorbance at 2.55 microns

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:48Z
Check the source to verify that this is the current version before use.

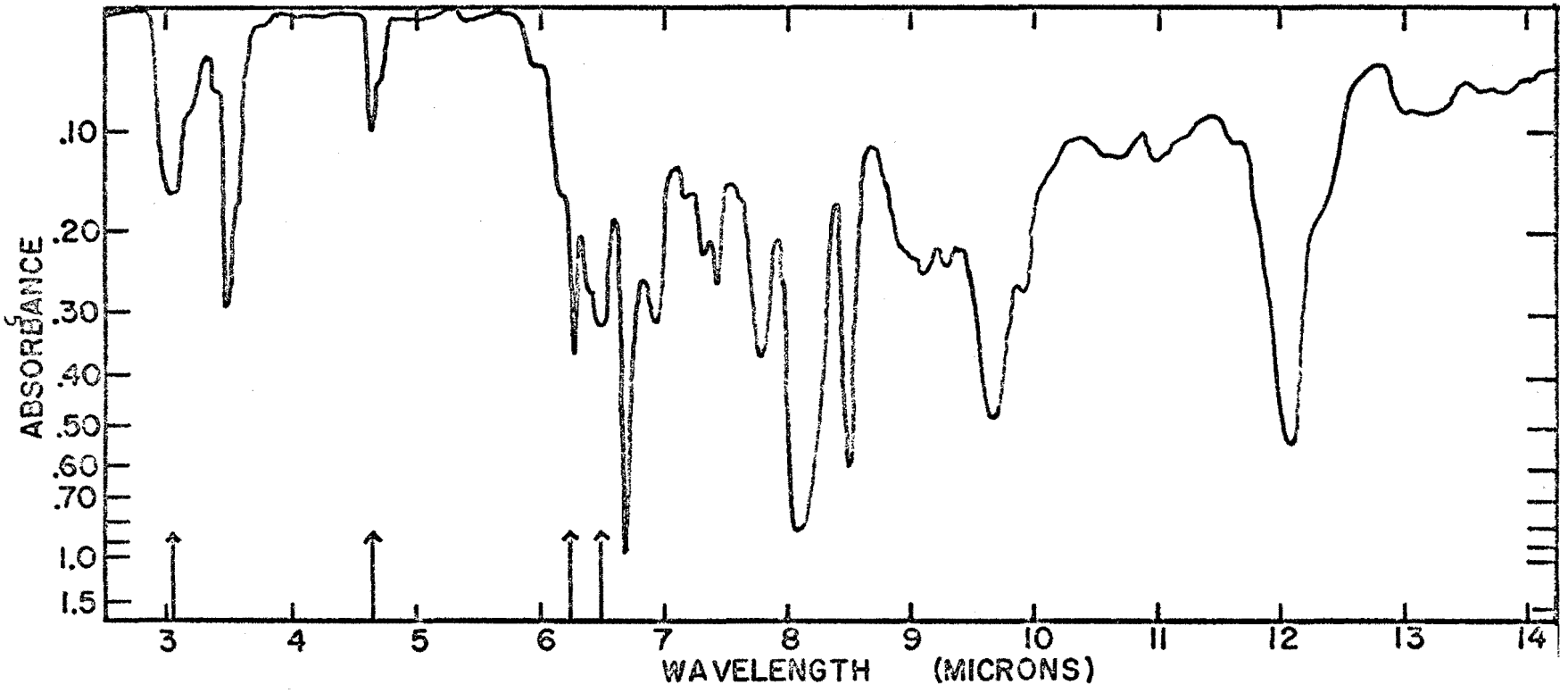


FIGURE I. Infrared spectrum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:48Z
Check the source to verify that this is the current version before use.

Table III. Absorbance ratios

Wavelength μ	$A_r \pm 0.03$ (absorbance ratios)
3.05	0.30
4.65	0.15
6.25	0.65
6.47	0.59

4.4.4 Viscosity, reduced. Reduce the package coating 1 to 1 by weight with a solvent of the following composition by weight: 3 parts diacetone alcohol, 3 parts butyl acetate, and 2 parts toluene. Determine the viscosity in accordance with method 4282 of Fed. Test Method Std. No. 141 for compliance with table I.

4.4.5 Drying time. Cast a film of the coating on a clean glass panel with a doctor blade having a clearance sufficient to give a dry film thickness between 0.0009 and 0.0011 inch after drying.

4.4.5.1 Set to touch. Determine set to touch in accordance with method 4061 of Fed. Test Method Std. No. 141 under referee conditions for compliance with table I.

4.4.5.2 Full hardness. Determine full hardness under referee conditions using a baking schedule of 200° ± 2°C. (392 ± 4°F.) for 70 minutes for compliance with table I. The film shall be considered to have reached full hardness when it is very difficult to remove with a knife blade.

4.4.6 Condition in container. Determine package condition in accordance with method 3011 of Fed. Test Method Std. No. 141 for compliance with 3.3.1.

4.4.7 Accelerated storage stability. Determine storage stability in accordance with method 3022 of Fed. Test Method Std. No. 141, except the test containers shall be stored for 7 days in a well ventilated convection oven maintained at 60° ± 2°C. (140° ± 4°F.). Determine viscosity in accordance with method 4271 and color in accordance with method 4248 of Fed. Test Method Std. No. 141 for compliance with 3.3.2.

4.4.8 Dipping properties. Reduce coating as in 4.4.4. Panels of AZ-31B magnesium conforming to QQ-M-44 shall be solvent cleaned in accordance with MIL-M-3171. Preheat panels to 200° ± 4°F. and immerse in the reduced coating for 30 seconds. Withdraw panels at a uniform rate of 10 ± 1 inches per minute. Air dry coated panels for 30 minutes and flash bake for 10 minutes at 392° ± 4°F. Cool panels to about 200°F. and repeat the above schedule for a total of 3 dips, except bake after the final dip for 70 minutes. Determine compliance with 3.3.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:48Z
Check the source to verify that this is the current version before use.

4.4.9 Spraying properties. Spray the reduced coating on panels cleaned as in 4.4.8 to give a dry film thickness between 0.0009 and 0.0011 inch after baking. The rate of application should be adjusted so as to give the required film thickness by two coats or passes of the spray gun. The passes shall be at right angles to each other. Air dry the coated panels for 30 minutes and bake at 392° ± 4°F. for 70 minutes. Determine compliance with 3.3.4.

4.4.10 Flexibility. Apply a 2 inch wide film of the packaged coating on tin plate panels cleaned in accordance with method 2012 of Fed. Test Method Std. No. 141 using the aliphatic naphtha-ethylene glycol monoethyl ether mixture. Use a film applicator that will give a dry film thickness between 0.0009 and 0.0011 inch after baking. Air dry coated panels for 30 minutes and bake as in 4.4.9. Condition panels for 24 hours under referee conditions, bend over a 1/8 inch mandrel and examine the area of the bend in accordance with method 6221 of Fed. Test Method Std. No. 141 for compliance with 3.3.5.

4.4.11 Salt spray resistance. Clean and treat AZ-31B magnesium panels conforming to QQ-M-44 in accordance with type III dichromate treatment of MIL-M-3171. Apply and bake the reduced coating as in 4.4.9. After baking score the film through to the metal substrate in a line about 1 inch from the top edge, running parallel to and about 1 inch from the long edge and ending about 1 inch from the bottom edge of the panel. Expose the panels to 20 percent salt spray for 240 hours in accordance with method 6061 of Fed. Test Method Std. No. 141. Upon removal, rinse panels free of soluble salt deposits and examine while wet and after a two hour recovery period for compliance with 3.3.6.

4.4.12 Adhesion (wet) tape test. Prepare a film of the coating as in 4.4.8 using panels of cast AZ-31B magnesium alloy. Determine the adhesion in accordance with method 6301 of Fed. Test Method Std. No. 141 modified as follows for compliance with 3.3.7:

(a) Scribe one-eighth inch cross-hatch pattern one inch square.
(b) Press the tape in firm contact with the coating and roll out all air bubbles by firm pressure of the thumb.

5. PREPARATION FOR DELIVERY

5.1 The coating shall be delivered in 1-quart or 1-gallon multiple friction top containers, in 5-gallon lug cover steel pails or in 55-gallon steel drums as specified (see 6.2). The coating shall be packaged level A or C; packed level A, B, or C as specified (see 6.2) in accordance with TT-P-143.

5.2 Additional markings. In addition to the marking required by TT-P-143 each container of the coating shall bear the following precautionary markings:

Caution. Contains highly volatile solvent of an extremely low flash point and extremely flammable nature. Observe all safety precautions in use. Avoid repeated or prolonged inhalation and skin contact.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:48Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The coating covered by this specification is intended primarily as a heat curable protective sealant applied to magnesium alloy castings used in military or other special equipment prior to application of other finishing coatings. The following outline of the recommended finishing procedure for magnesium castings is furnished for information only (for further details see MIL-M-46080(MR) "Magnesium Castings, Process for Anodic Cleaning and Surface Sealing of"):

(a) Anodic cleaning - fluoride anodizing.
(b) Removal of anodic film - chromic acid solution (15% CrO_3 by wt.).
(c) Chromate treatment - MIL-M-3171, type III.
(d) Seal with three dipcoats of coating conforming to this specification; MIL-C-46079(MR)
(e) Paint system - primer plus topcoat as specified.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Administrative provisions for inspection records (see 4.2).
(c) Size of cans or pails (see 5.1).
(d) Levels of packaging and packing required.
(e) The coating should be purchased by volume, the unit being one U.S. liquid gallon of 231 cubic inches at 20°C. (68°F.).

Custodian:
Army - MR

Preparing activity:
Army - MR

Review interest:
MU, MI, WC

Project No. 8010-A241

User interest:
ME, EL

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T04:48Z
Check the source to verify that this is the current version before use.